Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Lab intro

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

Capture-The-Flag (CTF) Competition!

URL: **https://play.webhack.it**

WebHackIT Scoreboard Rules Register Login

WebHackIT

WebHackIT is the Capture The Flag hacking contest of the **Data Privacy and Security course** at LUISS University. The competition is organised by **Emilio Coppa**, Lorenzo Ariemma, and Lorenzo Ralli.

Wait, what is a CTF?

A **Capture the Flag (CTF)** is a computer security competition. CTF contests are usually designed to serve as an educational exercise to give participants hands-on experience in the sort of attacks and protections found in the real world. There are two main styles of capture the flag competitions: jeopardy - like this competition - and attack/defense.

Jeopardy-style competitions usually involve multiple categories of problems, each of which contains a variety of questions that range from easy to more difficult ones. Teams or individuals attempt to earn the most points in the competition's time frame, but do not directly attack each other. CTFs often touch many aspects of information security, like cryptography, steganography, binary analysis and exploitation, reverse engineering, web security, mobile security and others. This competition will be focused on email security and web security (client and server-side).

Powered by {CTFORGE}

Wait, what is a CTF?

A **Capture the Flag (CTF)** is a computer security competition. CTF contests are usually designed to serve as an educational exercise to give participants hands-on experience in the sort of attacks and protections found in the real world. There are two main styles of Capture the Flag competitions: jeopardy (like this competition) and attack/defense.

Jeopardy-style competitions usually **involve multiple categories of problems, each of which contains a variety of questions that range from easy to difficult ones**. **Individuals** attempt to earn the most points in the competition's time frame, but **do not directly attack each other**. CTFs often touch many aspects of information security, like cryptography, steganography, binary analysis and exploitation, reverse engineering, web security, mobile security, and others.

CTFs are quite common in cybersecurity

Check out: **https://ctftime.org/**

CTFTIME | CTFs | Upcoming | Archive | Calendar | Teams | FAQ | Contact us | About | Sign in

Home / CTFs / Events / Upcoming

CTF Events

All | Now running | Upcoming | Archive | Format | Location | Restrictions | 2026

Name	Date	Format	Location	Weight	Notes	
BITSCTF 2026	06 Feb., 05:30 UTC — 08 Feb. 2026, 05:30 UTC	Jeopardy	On-line	25,54		**191** teams will participate
Pragyan CTF 2026	06 Feb., 13:00 UTC — 08 Feb. 2026, 13:00 UTC	Jeopardy	On-line	0,00		**56** teams will participate
LA CTF 2026	07 Feb., 02:00 UTC — 09 Feb. 2026, 02:00 UTC	Jeopardy	On-line	53,58		**115** teams will participate
Nullcon Goa HackIM 2026 CTF	07 Feb., 08:30 UTC — 08 Feb. 2026, 08:30 UTC	Jeopardy	On-line	24,72		**56** teams will participate
DUCKERZ CTF 2026	07 Feb., 09:00 UTC — 08 Feb. 2026, 09:00 UTC	Jeopardy	On-line	0,00		**39** teams will participate

Some are organized by big IT companies…

Google CTF:

Academic CTFs

One of the first university CTFs was **iCTF** by UC Santa Barbara:

iCTF: the International Capture The Flag Competition

The International Capture The Flag ("iCTF") is a distributed, wide-area security exercise, which aims to test the security skills of the participants. The iCTF is one of the world's largest and longest-running educational hacking competitions and has the goal of educating students about security-related topics in a live setting.

The Upcoming iCTF

The next iCTF is scheduled in March 9-13, 2026. Note that the 2025 iCTF was originally scheduled for the beginning of December 2025, but due to unforeseen circumstances, we had to postpone it.

Until then, teams can practice using the challenges from the 2023 and 2024 editions of the competition.

Join the discord channel here and monitor the channel for announcements and training materials.

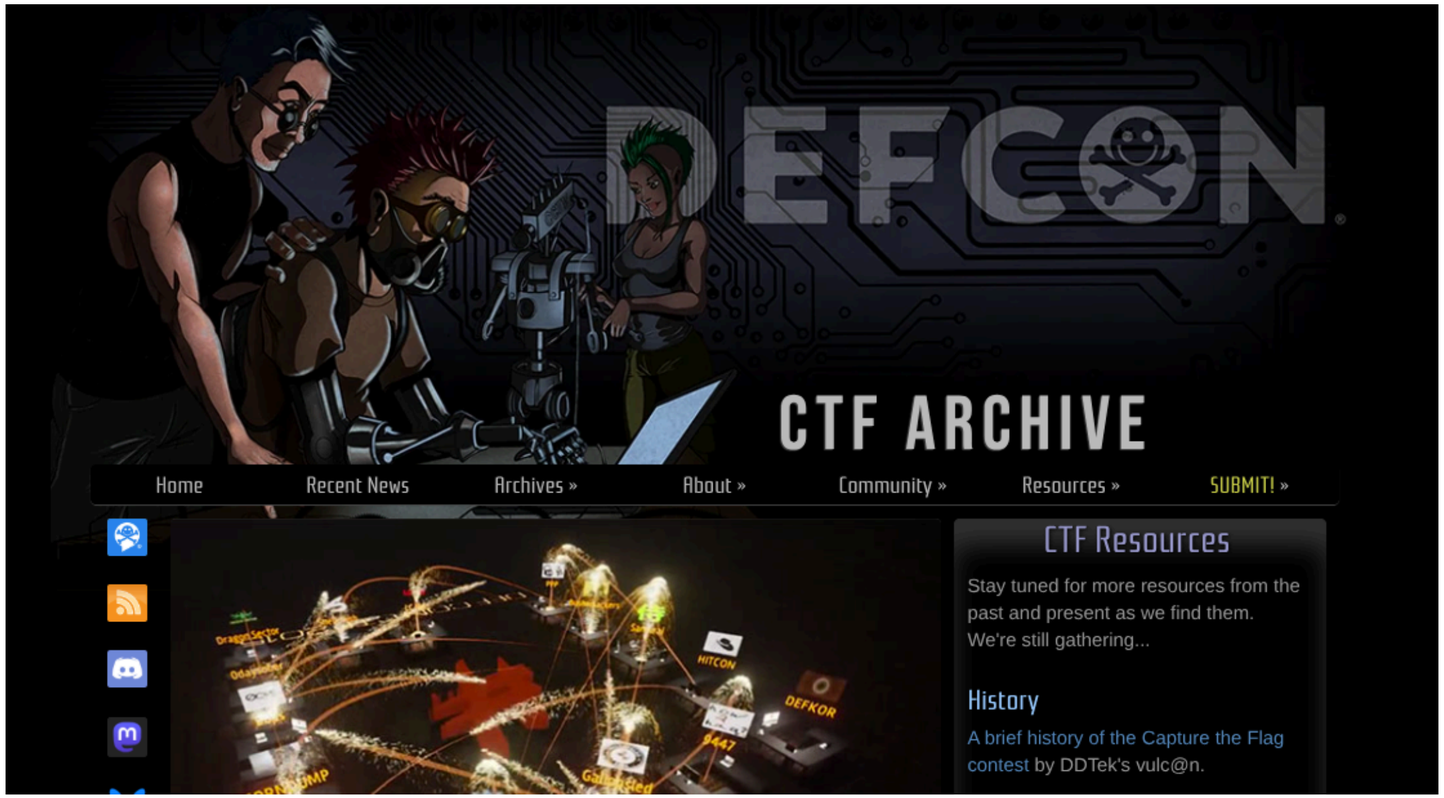
You can register a team here.

The competition is sponsored by the ACTION NSF AI Institute, and is organized by Shellphish.

History and Background

The iCTF evolved from several security "live exercises" that were carried out locally by Prof. Giovanni Vigna at UC Santa Barbara, in 2001 and 2002.

DEFCON CTF: the most iconic underground CTF



CTF as a training tool: learn by ~~doing~~ hacking

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/cose

ELSEVIER

Computers & Security

TC 11 Briefing Papers

Cybersecurity knowledge and skills taught in capture the flag challenges

Valdemar Švábenský, Pavel Čeleda, Jan Vykopal, Silvia Brišáková*

Masaryk University, Brno, Czech Republic

Capture the Flag as Cyber Security Introduction

Lucas McDaniel
University of Alaska Fairbanks
lamcdaniel@alaska.edu

Erik Talvi
University of Alaska Fairbanks
etalvi@alaska.edu

Brian Hay
University of Alaska Fairbanks
brian.hay@alaska.edu

Abstract

Introducing technical concepts to students with little to no technical background can be a challenging task for any teacher to achieve. The concept of gamification has been introduced recently as a method to motivate students by taking a variety of techniques found in popular games, and adding them into educational modules. Extending from this notion, it has been found that capture the flag (CTF) style competitions are a successful way to introduce students to a variety of technical concepts within the standard computer science curriculum.

memorizing answers or shortcuts to get answers to bypass the lectures and materials that they perceive as boring [9]. This means that they will not only fail to fully grasp the topic at hand, but will quickly forget it after the course or even the lecture has finished.

Therefore many instructors have moved to using hands-on learning techniques so that students are able to explore the application of security concepts in a real world environment. In the past this has been done with labs and tutorials including those that are instructor-guided or others that are exploratory in nature, but recently there has been a trend to gamify the environment.

Earn points by solving challenges

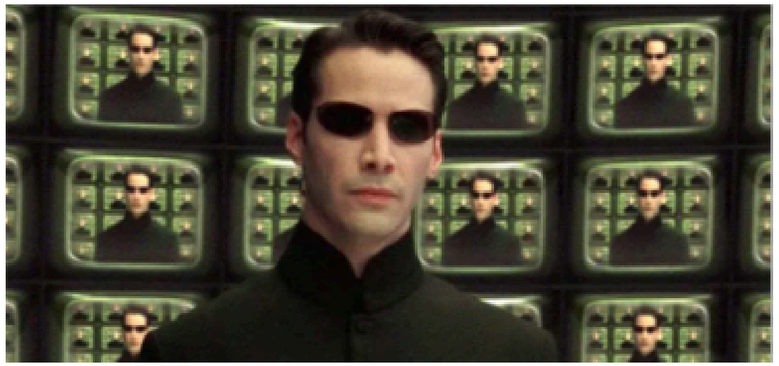
Each challenge:

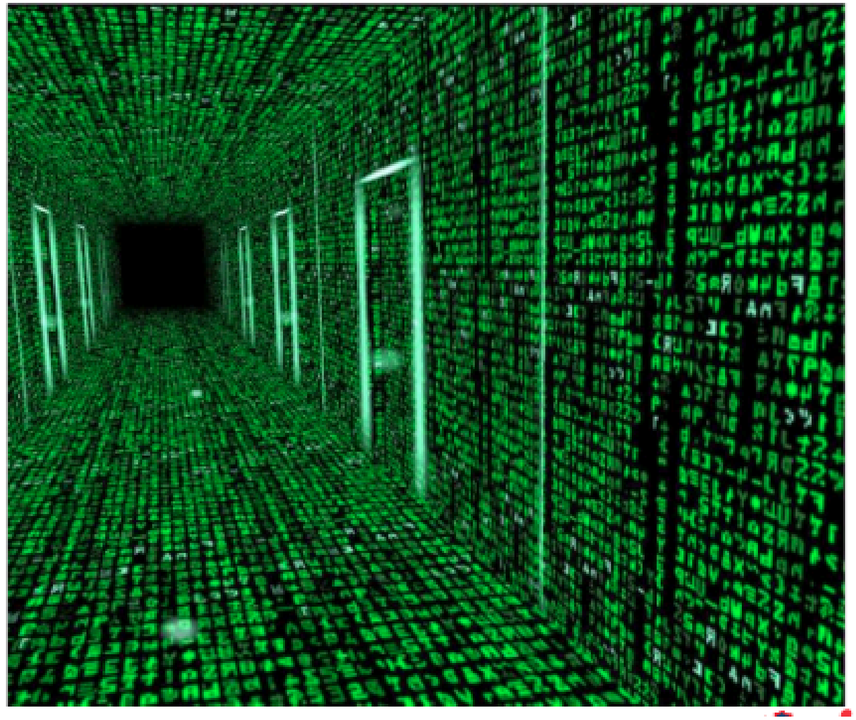
- It is a web application (in WebHackIT!)
- It is designed to reproduce a real-world problem inside an isolated environment
- It may simplify some assumptions and details to help understand the goal
- In some cases, you have to show knowledge of some topics... they are just exercises
- While in other cases, you have to find and then exploit a vulnerability!

How?

- The solution will depend on the specific challenge
- During the course, we will show you how to solve some challenges
- ...some of these challenges have been inspired by other past CTFs
- We will show you some tools that can help you

You think it will be like this…





It will be like this…

How do you show that you have solved the challenge?

- Each challenge contains a secret string (which is hard to guess!)
- This is called a flag... hence the name Capture-The-Flag!
- A flag can be obtained after successfully performing some “tasks”, or it can be the result of an attack (e.g., you can log in as an admin)
- Easy to recognize since the format will be: `WIT{<random chars>}` [hard to guess!]
- Just insert the flag on the web portal to get the points!

How the score is calculated?

How the score is calculated?

For the game and just for fun:

- each challenge is initially worth 500 points
- the points are gradually reduced the more people solve the challenge
- hence, an easy challenge will be worth less than a hard challenge

How the score is calculated?

For the game and just for fun:

- each challenge is initially worth 500 points
- the points are gradually reduced the more people solve the challenge
- hence, an easy challenge will be worth less than a hard challenge

For the exam:

- the score is given by:

$$30 \cdot \frac{\text{number of challenges solved by a student}}{\text{total number of challenges presented in class}}$$

- other challenges, not presented in class, may give bonus points

Can we solve the challenges in groups?

No, the competition is played individually by each student.

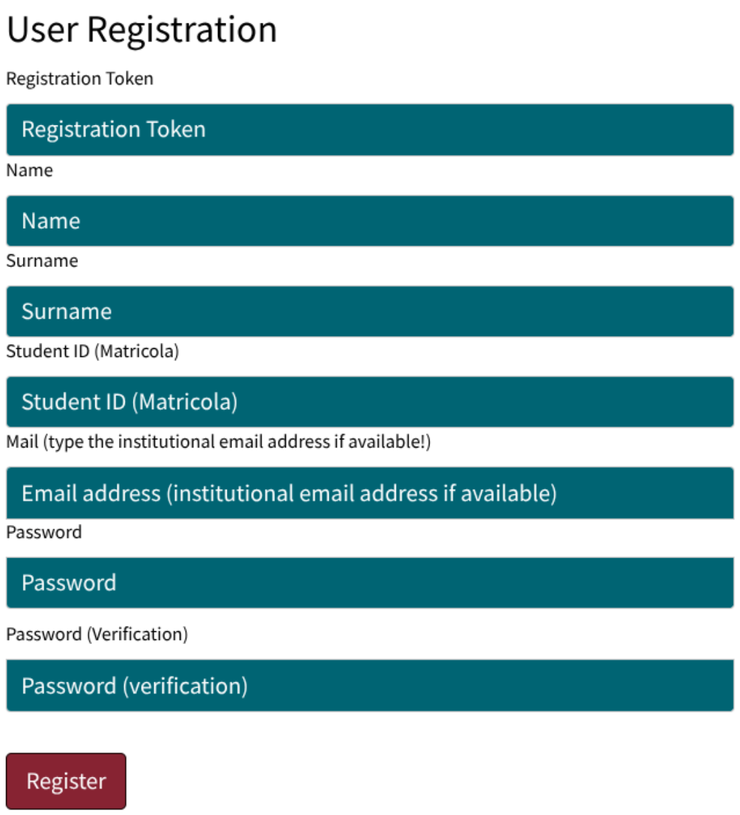
You have to play with your account on your laptop.

You have to actually perform all the steps on your own.

Can I solve the challenge after the lecture?

You can submit the flag to the CTF platform within the day of the lecture*. After that, the platform will still accept the flag, but I may ignore your submission at the end of the course.

* If we notice that students stop following the lectures, we will remove this flexibility.

First challenge? Register an account!

- URL: **https://play.webhack.it/register**
- Registration token: **`webhackit_2526`**